

STG17 · TECHNICAL WORKSHOP · AFRICA AFTER DARK

Lumina: an AI agent to read Africa at night

Architecture, workflow and toolbox of the “Night Lights” agent

Groq · Gemini · Ollama

Google Earth Engine

21 tools

Colab · Kaggle · Local



Six questions to understand the agent

01

Why an agent?

What is an agent for, and how does it differ from a chatbot?

02

Architecture

How do the four layers fit together?

03

How it works

What happens between the question and the answer?

04

Toolbox

What are the 21 tools and what do they compute?

05

Data

Which data, over which periods, with which pitfalls?

06

Reliability & use

How do we ensure reliable answers and use them?

From ISO code to indicator, no manual pipeline

Classic pipeline

- 1 Download the VIIRS rasters
- 2 Clip to the country extent
- 3 Zonal statistics by region
- 4 Series, trends, tests
- 5 Charts and maps
- 6 Write the note and its limitations

Several days · GIS + Python skills

With the agent



“Which regions progressed most since 2015?”



Structured note + map + chart + trace, in a few minutes

21

tools

3

LLM providers

5

data sources

An agent = an LLM + tools + a loop



The LLM

Understands the question, plans, chooses the tools and writes the final note.



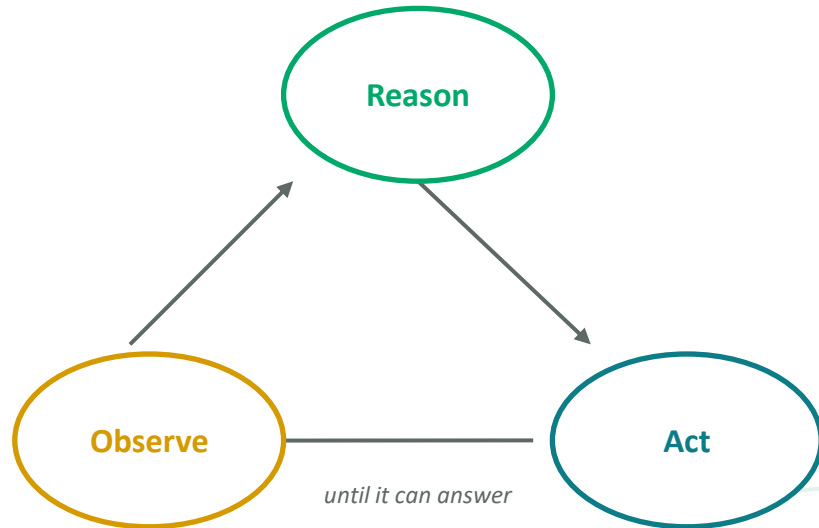
The tools

21 Python functions computing on real data (Earth Engine, World Bank).



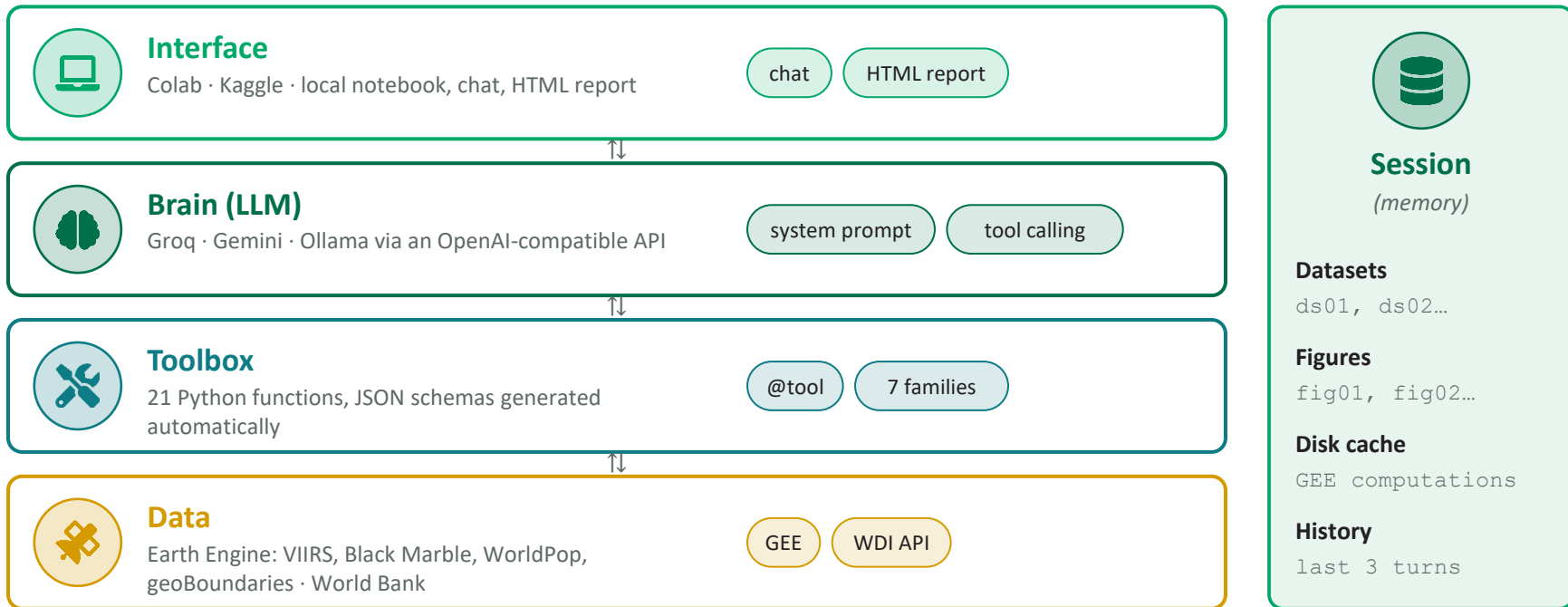
The loop

Repeats reason → act → observe until it can answer (12 steps maximum).

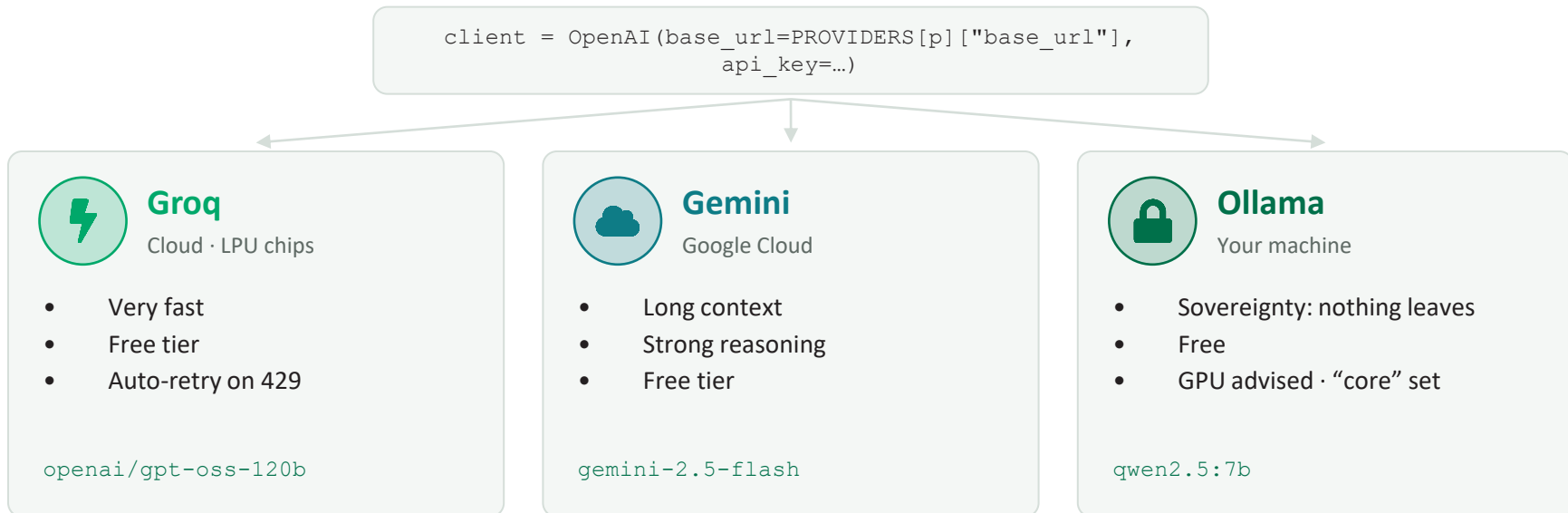


Chatbot: answers from memory, risk of hallucination. **Agent:** every number comes from a traced, verifiable tool.

Four layers, one session memory

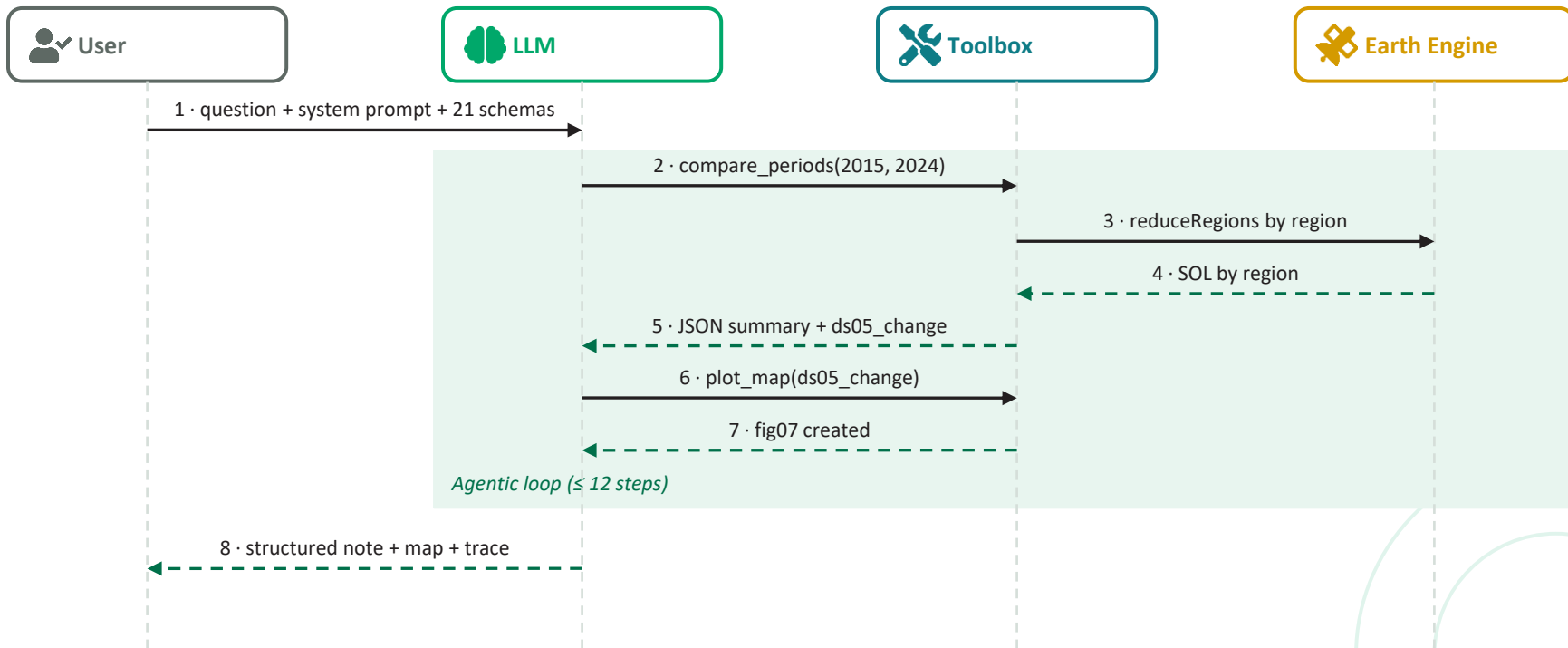


One interface, three interchangeable brains



Model checked at start-up: if it has been retired from the catalogue, automatic fallback to the first available model in a preference list.

The loop step by step: a real example



Anatomy of a tool: the signature becomes a contract

① Python function

```
@tool("Spatial analysis")
def compare_periods(
    year_start: int,
    year_end: int,
    level: int = 1,
    iso3: str | None = None):
    """Compare two years
    by administrative unit:
    change in SOL, CAGR...
    Args:
        year_start: Start year"""
```

Types → JSON Schema

② JSON schema (seen before)

```
{
  "name": "compare_periods",
  "description": "Compare two
  years by unit...",
  "parameters": {
    "year_start": {"type": "integer"},
    "year_end": {"type": "integer"},
    "level": {"type": "integer",
      "description": "... (default: 1)"}
  },
  "required": ["year_start",
    "year_end"]}

```

Docstring → description

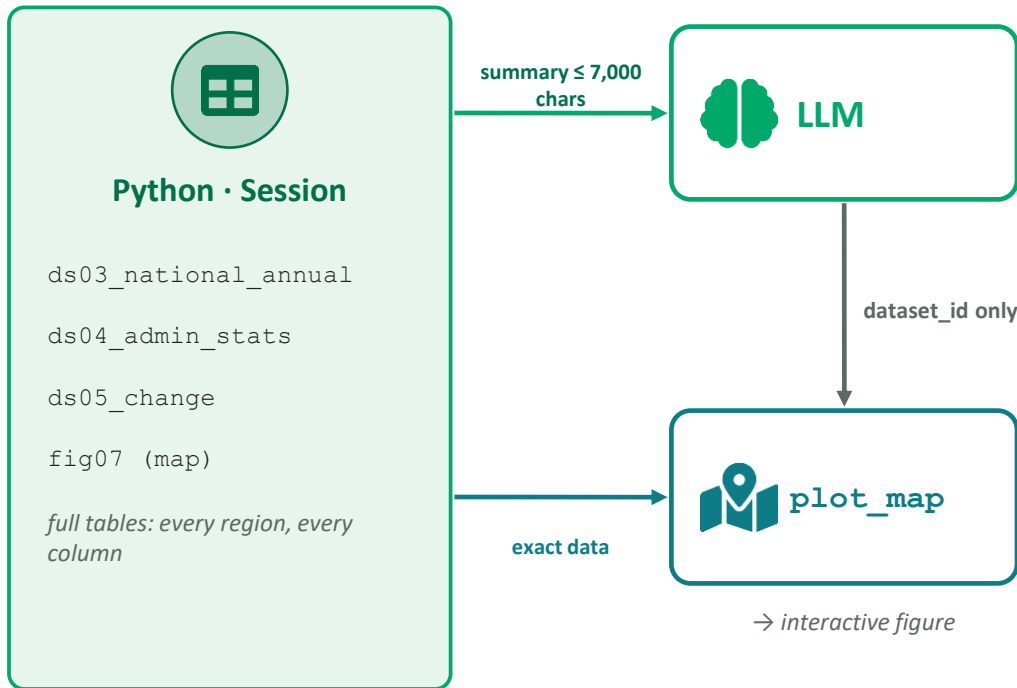
③ Result (seen after)

```
{
  "ok": true,
  "dataset_id": "ds05_change",
  "national_sol_change_pct": 42.3,
  "top_gainers": [ ...5 rows... ],
  "top_decliners": [ ...5 rows... ],
  "convergence_test": {
    "reading": "convergence..."},
  "notes": ["Small units:
    % highly volatile"]}

```

Summary + dataset_id, not the table

The LLM sees summaries, Python keeps the data



No invented values

Charts re-read the exact data.



Short context

Fewer tokens: faster and cheaper.



Reuse

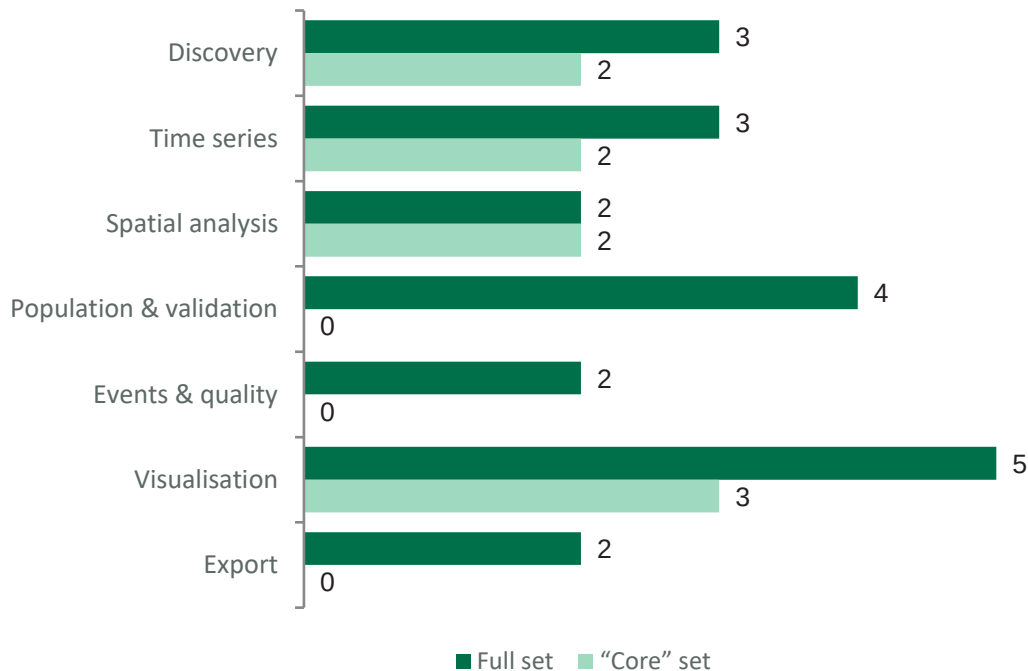
One dataset feeds several charts, with no recomputation.



Disk cache

An identical Earth Engine computation is never re-run.

21 tools in 7 families



Source: STG17 notebook, count of tools declared with @tool.

21

tools in the full set

9

"core" tools for small models:
TOOLSET = "core"

Analysis tools: from scoping to validation



Discovery

```
get_country_profile  
get_data_availability  
list_admin_units
```

Available years, regions, area



Time series

```
get_national_timeseries  
get_admin_timeseries  
detect_anomalies
```

Trend, CAGR, anomalies vs clouds



Spatial analysis

```
get_admin_stats  
compare_periods
```

Ranking, spatial Gini, convergence



Population & validation

```
get_population_light_stats  
get_worldbank_indicator  
correlate_ntl_with_indicator  
compare_countries
```

Unlit population, proxy vs GDP, peers



Events & quality

```
assess_event_impact  
get_quality_report
```

Black Marble before/after, clouds, flares



Export

```
export_dataset  
list_session_datasets
```

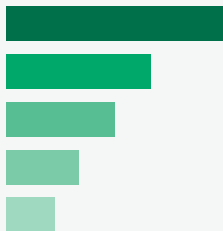
CSV / Excel, reuse of results

Five visual tools, fed by a dataset_id



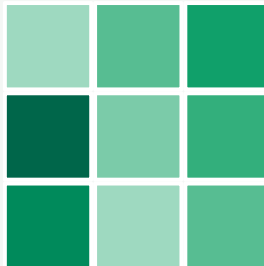
`plot_timeseries`

Change, trend, anomalies



`plot_ranking`

Ranking of regions



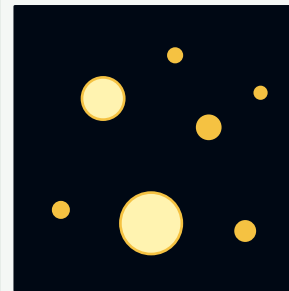
`plot_map`

Choropleth map of regions



`plot_scatter`

Validation: SOL vs GDP



`show_night_image`

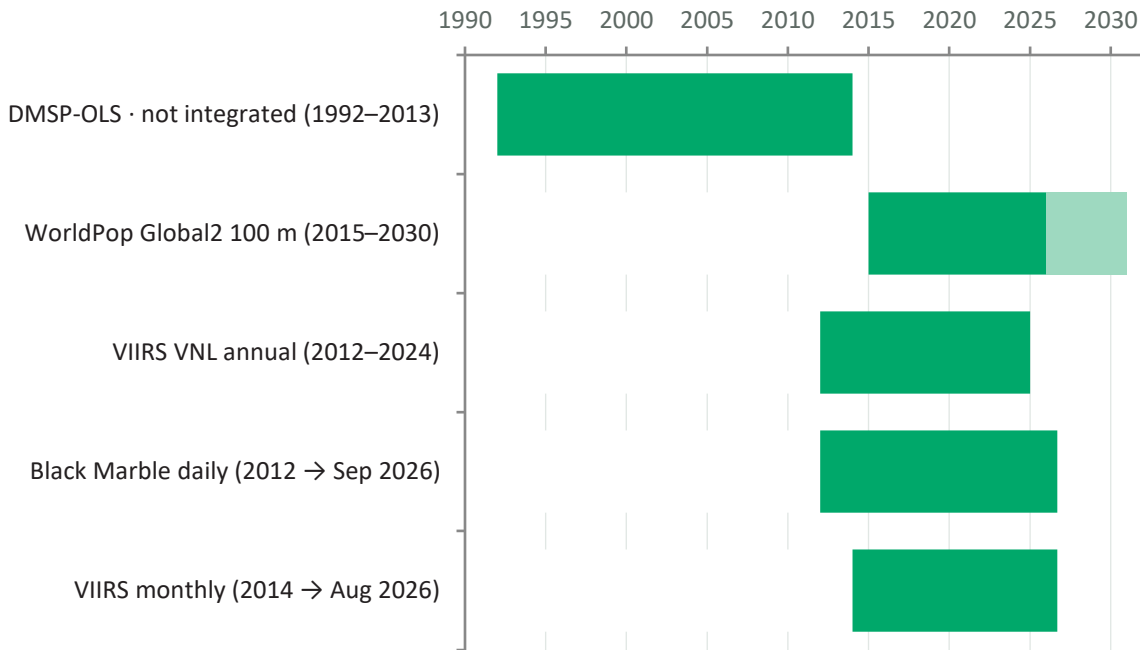
The country at night

Schematic previews.



The LLM passes a dataset_id, never numbers: it cannot invent a plotted value.

Which periods are covered?



Annual

up to 2024: the agent's default year



2025–2026

via monthly composites, never mixing two products



Population

WorldPop Global2 for the same year as the lights

Light green: WorldPop projections (2026–2030). Sources: Earth Engine catalogue (1 Oct 2026); WorldPop Global2 R2025A; NOAA/EOG.

Four indicators, six pitfalls to know

SOL

Sum of radiances

Proxy for total activity

Rad.

Mean radiance

$\text{SOL} \div \text{number of pixels}$

km²

Lit area

Pixels $> 0.5 \text{ nW/cm}^2/\text{sr}$

Pop.

Unlit population

WorldPop below threshold

Artefact



Cloud cover



Seasonality



Gas flares



Urban halo (blooming)



Rural noise



Product break

Countermeasure in the agent

anomalies cross-checked with coverage

deseasonalised before detection

percentiles + producer-country alert

threshold + caution at ADM2

$0.5 \text{ nW/cm}^2/\text{sr}$ threshold

a single source per series

Six safeguards for a reliable answer



1

No number without a tool

Forbidden by the system prompt; charts re-read the exact data.



2

Errors sent back

A tool error becomes a clear message: the LLM fixes its arguments.



3

Clouds checked

A drop is interpreted only after checking cloud coverage.



4

Mandatory limitations

Every answer has a “Limitations and caveats” section.



5

Audit trail

Tools, exact arguments, durations and results are all logged.

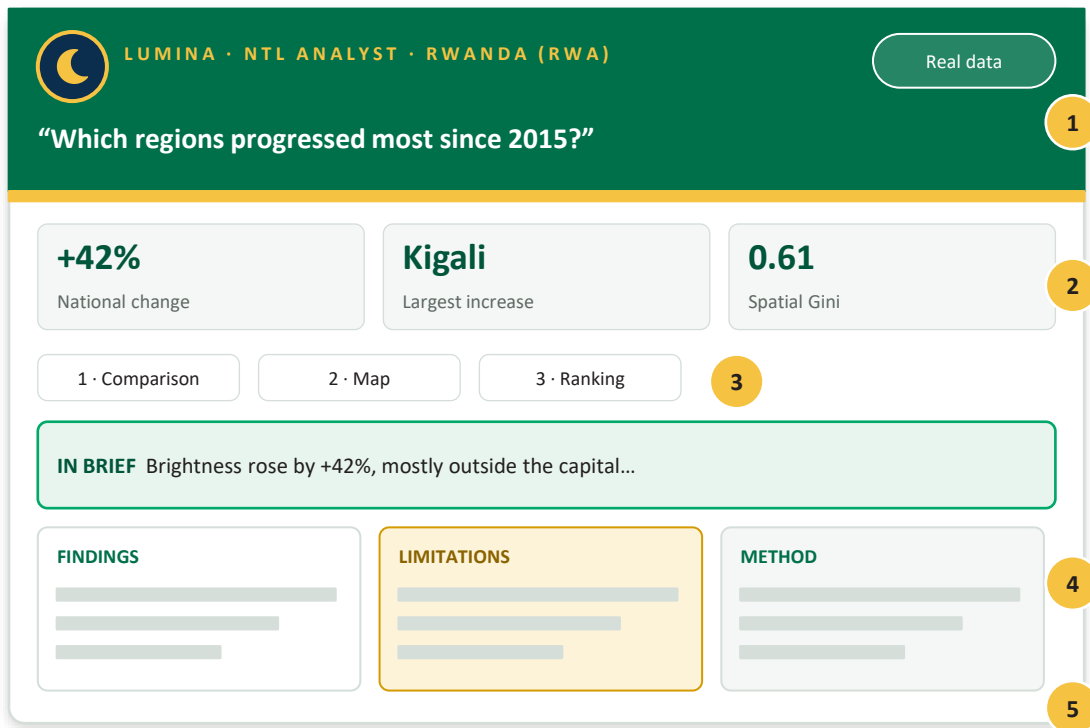


6

Evaluation

Ground truth computed in Python, compared with the answer.

What the user sees: an analytical note



- 1 Header**
Question, country, model, duration, data status
- 2 Key indicators**
Computed by the tools, never by the LLM
- 3 Workflow**
The tools called, in order, with their status
- 4 Structured note**
In brief · Findings · Interpretation · Limitations · Method
- 5 Figures + trace**
Numbered charts, expandable audit trail

Schematic mockup; illustrative values.

Get started in three steps



1 · Keys

GROQ_API_KEY or GEMINI_API_KEY ·
GEE_PROJECT in Secrets



2 · Settings

COUNTRY_ISO3 · LLM_PROVIDER ·
DATA_BACKEND · TOOLSET



3 · Ask

Run all, then ask("...") or use the chat

Google Colab

Secrets  · one-click Earth Engine
authorisation

Kaggle

Enable Internet · GEE service account advised

Local

.env file · earthengine authenticate



No Earth Engine account yet? DATA_BACKEND = "demo": synthetic data, flagged everywhere, for learning. Never quote these numbers.

KEY TAKEAWAYS

A reliable agent starts with a good toolbox



The LLM orchestrates, the tools compute



A clear docstring = a well-chosen tool



Python keeps the data, the LLM sees summaries



Every answer is traced and validated by a human

YOUR TURN

Add your own tool with
`@tool: urban expansion,`
newly lit areas, comparison
with the census...

Sources and references



Data

- Elvidge C.D. et al. (2021). Annual time series of global VIIRS nighttime lights. Remote Sensing 13(5).
- Román M.O. et al. (2018). NASA's Black Marble nighttime lights product suite. Remote Sensing of Environment 210.
- Runfola D. et al. (2020). geoBoundaries. PLoS ONE 15(4).
- WorldPop (2025). Global2 R2025A: global 100 m population, 2015–2030.
- World Bank. World Development Indicators, API v2.



Platforms and APIs

- Earth Engine catalogue: NOAA/VIIRS/DNB/ANNUAL_V22, MONTHLY_V1/VCMSLCFG, NASA/VIIRS/002/VNP46A2
- GEE community catalogue: projects/sat-io/open-datasets/WORLDPOP/pop
- Groq: console.groq.com/docs/openai
- Gemini: ai.google.dev/gemini-api/docs/openai
- Ollama: github.com/ollama/ollama (docs/openai.md)
- World Bank: worldbank.github.io/OpenNightLights